

Revealing Hidden Performance Issues with Extended Events

Jeff Iannucci

Download all slides and scripts



<https://github.com/desertdba/Presentations>

Who in the world is Jeff Iannucci?



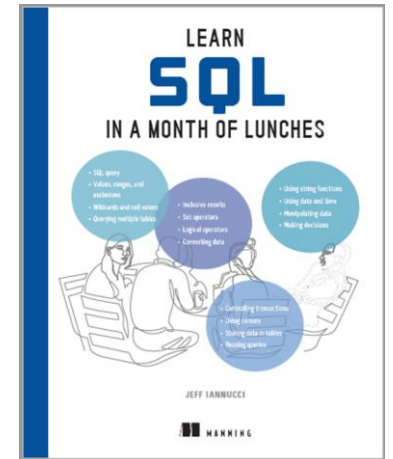
Consultant at Straight Path Solutions



Content Author at Pluralsight



Author of “Learn SQL in a Month of Lunches”



XE

XML



Select Events To Capture

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Select the events you want to capture from the event library.

Event library:

Search Events

in Event names only

Name ^	Category v	Channel v	Package v
query_plan_profile	execution	Analytic	sqlserver
query_post_compilation_showplan	execution	Analytic	sqlserver
query_post_execution_plan_profile	execution	Analytic	sqlserver
query_post_execution_showplan	execution	Analytic	sqlserver
query_post_execution_stats	execution	Analytic	sqlserver
query_pre_execution_showplan	execution	Analytic	sqlserver
query_store_async_queue_diagnostics	query_store	Operational	qds
query_store_auto_enable_failure	query_store	Operational	qds
query_store_auto_enable_on_create	query_store	Operational	qds
query_store_background_cleanup_task_failed	query_store	Operational	qds
query_store_background_task_abort_stats	query_store	Operational	qds
query_store_background_task_creation_failed	query_store	Operational	qds
query_store_background_task_initialization_failed	query_store	Operational	qds
query_store_background_task_persist_failed	query_store	Operational	qds
query_store_background_task_persist_finished	query_store	Operational	qds

query_parameterization_data

Fired on compile for every query relevant for determining if forced parameterization would be useful for this database.

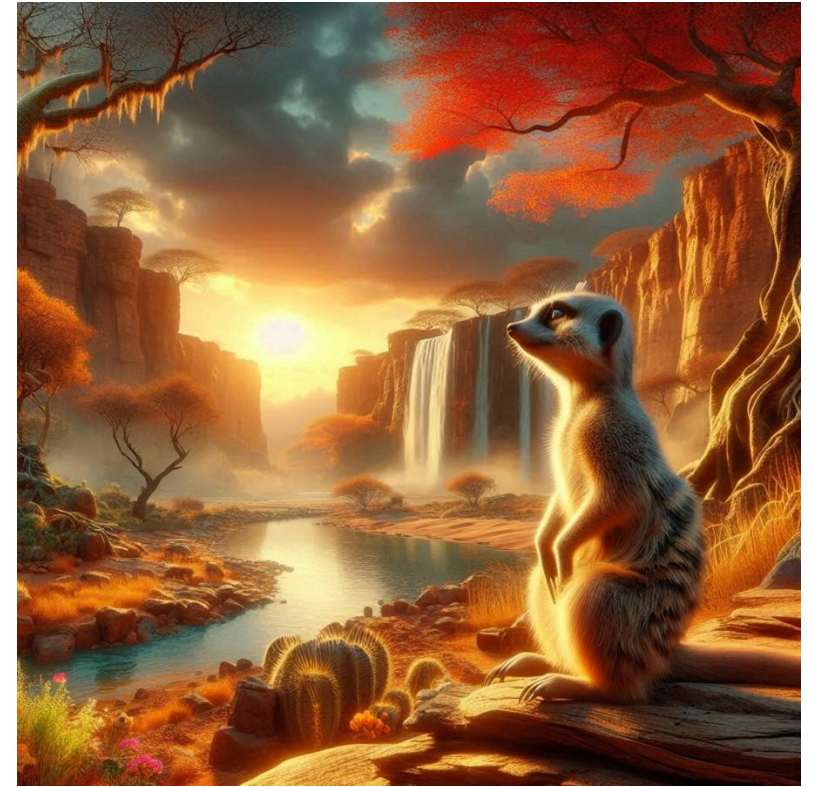
Event Fields ^

Event Fields	Description
cached_plan_size	Cached plan size (KB).
compile_code	Compile code describin...
compile_cpu_time	Query compile cpu time.
compile_duration	Query compile elapsed ...
compile_memory	Memory (KB) used durin...
database_id	ID of the database
has_external_extraction	Indicates whether this s...
has_literals	Indicates whether quer...
is_cached	Indicates whether this s...
is_parameterizable	Indicates whether quer...
is_query_variant	Indicates whether this s...
is_recompiled	Indicates whether this s...
mgf_persistence_used_...	Indicates status of appl...
parameterized_values_...	Number of parameters i...
query_hash	Query hash, which repr...



What is an Extended Event?

“A SQL Server Extended Event is a **lightweight** performance **monitoring** system that allows users to collect and analyze detailed information about database engine events.”



Goals

Which new Events find performance issues

Which new Events aren't as helpful

What painless ways we can view Event info



Parts of an Extended Event

EVENT

What are we watching for?

ACTION

What information do you want?

TARGET

Where do you want the info?



Permissions to create an Extended Event

~~SQL Server 2008 and 2008 R2:~~

~~CONTROL SERVER~~

SQL Server 2012 to 2019:

ALTER ANY EVENT SESSION

SQL Server 2022 and higher:

CREATE/DROP ANY EVENT SESSION



Permissions to view an Extended Event

SQL Server 2008 to 2019:

VIEW SERVER STATE

SQL Server 2022 and higher:

VIEW SERVER PERFORMANCE STATE



What's new with Extended Events?

Query antipatterns

`query_antipattern`

Aborted queries

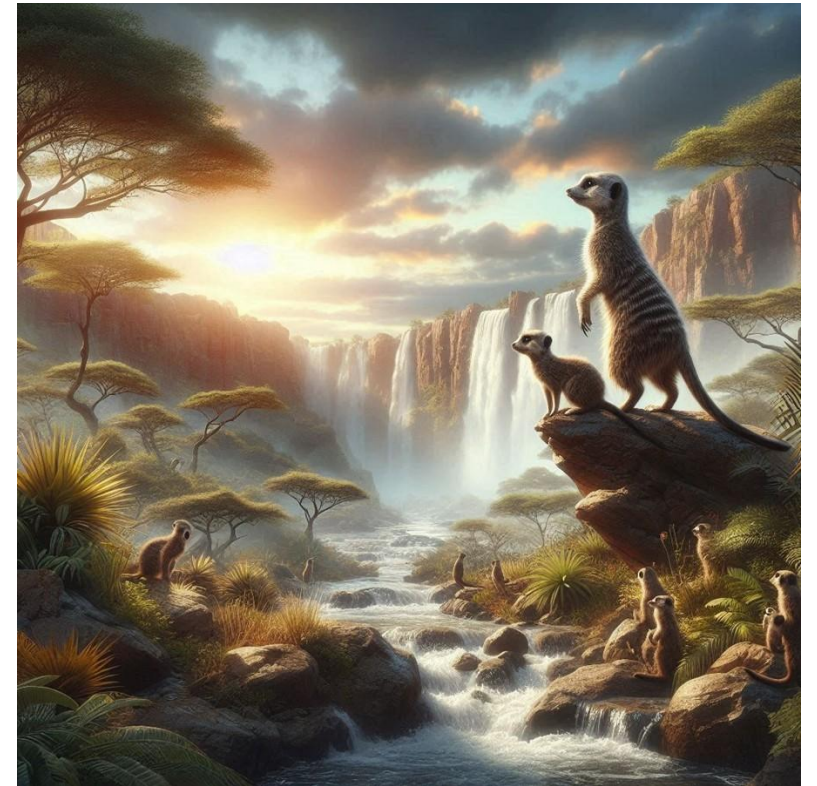
`query_abort`

Spills to tempdb

`spills_to_tempdb`

Measuring extended event overhead

`(sys.dm_xe_session_events)`





DEMO TIME!

The basics

Long-running queries

`sql_statement_completed`

Blocking

`blocked_process_report`

`(sp_HumanEventsBlockViewer)`

Deadlocks

`lock_deadlock`

`(system_health XE)`

`(sp_BlitzLock)`



Failed and aborted queries

Failed queries

error_reported

Aborted queries

query_abort

Aborted queries – old school

sql_batch_completed

rpc_completed

sp_statement_completed



Antipatterns

Type Convert Preventing Seek

query_antipattern (...mostly...)

Large Number of OR in Predicate

query_antipattern (...sort of...)

Large IN clause

query_antipattern (in 2025)

Non-optimal use of OR

query_antipattern (in 2025)



Monitoring tempdb

Growth

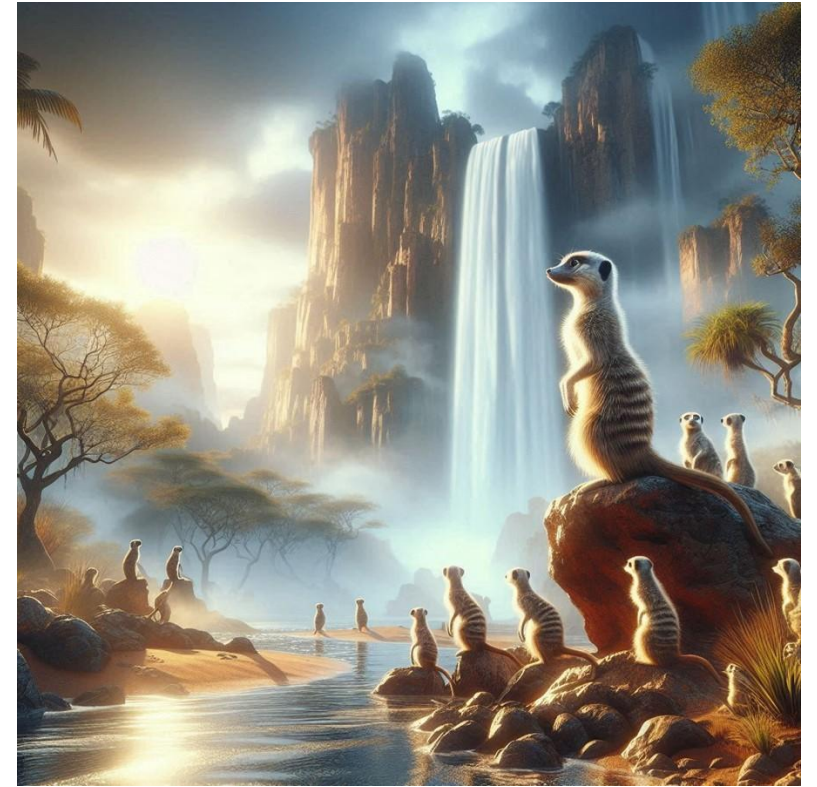
database_file_size_change

database_log_file_used_size_change

Spills

spills_to_tempdb (-_(\ツ)_/~)

sql_statement_completed



Measuring extended events

Impact

Trace flag 9708

sys.dm_xe_session_events



BONUS!

sp_HumanEvents

Blocking

Query performance

Compiles

Recompiles

Wait statistics



https://erikdarling.com/sp_humanevents/

ANOTHER BONUS!!

XE Smart Target

Uses XE streaming API to connect

Allows for responses:

- Write to CSV file
- Email
- Execute T-SQL
- Replay
- Write to a table



<https://github.com/spaghettidba/XESmartTarget>

MOAR BONUS!!!

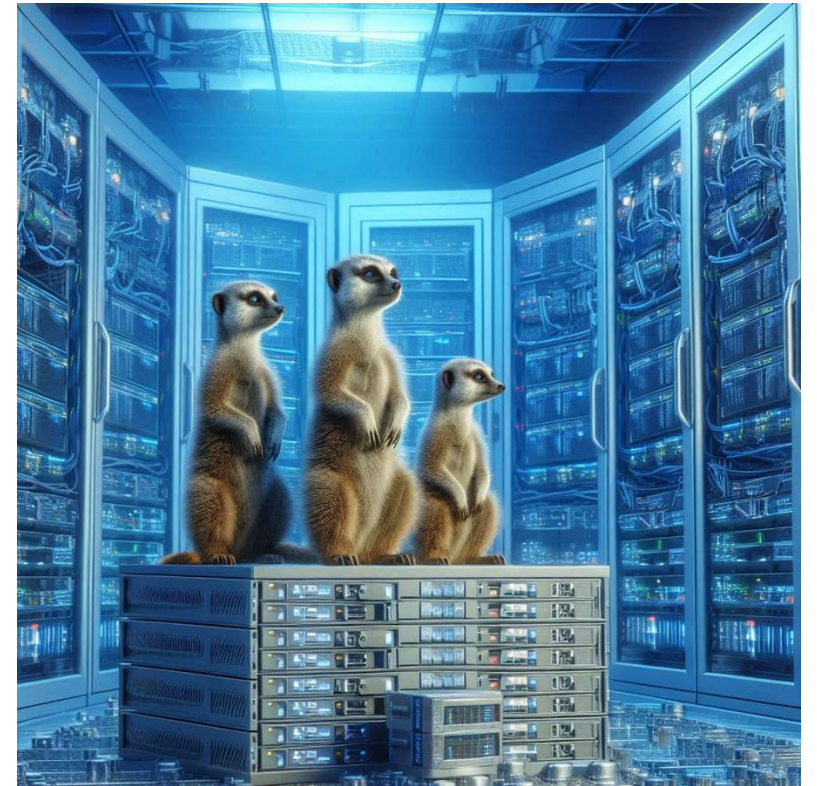
dbatools.io

Get-DbaXESession

Read-Dba-XEFile

Get-DbaXESmartTarget

...and 25 more cmdlets!



<https://dbatools.io/commands/#Traces>



WARNING!

Please do not start all the ~~meerkats~~ extended events at the same time.

Thank you!!



<https://github.com/desertdba/Presentations>



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